When Do Demonstrations Teach the Wrong Lesson? Spurious Label Statistics Barely Move Modern Instruction-Tuned LLMs

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Abstract

Few-shot prompts are often assembled by hand, so their demonstrations can carry spurious statistics: one label may dominate, sit at the end of the prompt, or come with longer inputs. Studies of GPT-3-era base models found that such statistics pull predictions strongly toward the favoured label. We ask how large these effects are in current instruction-tuned models and how they change with scale. We build a fully paired benchmark of 10,000 prompts per model over four classification tasks. It covers four manipulations of a 10-shot demonstration set: frequency skew (80/20), recency, primacy, and label–length correlation. The favoured label is rotated through all labels, so intrinsic label preferences cancel. We score eight local models: Qwen2.5-Instruct from 0.5B to 14B, a Qwen2.5-1.5B base model, and two Llama-3 models. Every bias is far smaller than hypothesised. Frequency skew is the largest effect in small models (+4.3 to +16.9 percentage points) and falls to about zero at 8–14B (-3.7 points per tenfold increase in parameters). Recency does not pull predictions toward the last label; from 3B upward it produces a small, significant *anti*-recency effect (-1.7 to -3.6 points). Label–length correlation has no reliable effect, and calibration error does not change reliably. Large shifts appear only under extreme manipulations, such as all demonstrations sharing one label (up to +36 points), and mostly on the ambiguous paraphrase task. For models of 3B and above, re-drawing a balanced demonstration set flips more predictions (4–8%) than any systematic bias we measure.

1 Introduction

Few-shot prompting is the default way to turn a large language model (LLM) into a classifier: paste a handful of labelled examples into the prompt, then ask for the label of a new input [Brown et al., 2020]. The demonstrations are usually picked by hand. As a result, they can carry statistics that have nothing to do with the task. Most demonstrations may share one label. The last few may share a label. One label may come with longer inputs than the others. If the model picks up these statistics, a deployed classifier can quietly change its behaviour depending on which examples someone happened to paste.

Early evidence suggested this is a serious problem. Zhao et al. [2021] showed that GPT-3 is pulled toward the majority label and toward the label of the last demonstrations, and that this makes few-shot accuracy unstable. Lu et al. [2022] showed that the order of demonstrations alone can move accuracy from near chance to near state of the art. Fei et al. [2023] identified a further domain-label bias. These findings motivated a family of calibration methods [Zhao et al., 2021, Fei et al., 2023, Zhou et al., 2024].

Do these biases still hold? Most of this evidence comes from GPT-3-era base models, and most studies vary one factor at a time. Practitioners now mostly use instruction-tuned models [Ouyang et al., 2022] from 1B to tens of billions of parameters. For these models we lack a side-by-side

39th Conference on Neural Information Processing Systems (NeurIPS 2025).

answer to two simple questions. First, which spurious demonstration statistics matter, and by how much? Second, does the answer change with model scale? A common intuition, which we turn into explicit hypotheses in section 3, is that frequency skew causes the largest bias (15–25 percentage points), recency comes second (8–15 points), label–length correlation causes a moderate bias that scale cannot fix, larger models correct skew but amplify recency, and calibration gets worse under every bias.

Our approach. We test these claims with one paired protocol. We apply four manipulations to a 10-shot demonstration set: frequency skew, recency, primacy (label position), and label–length correlation (table 1). We compare each against a balanced CLEAN set on SST-2, RTE, AG NEWS and MRPC. The label favoured by the manipulation, the *target* T, is rotated through every label. A model's intrinsic preference for one label therefore cancels, and the shift toward T measures the bias directly. Every model sees the same 10,000 prompts. We score labels by their exact next-token probabilities, which removes parsing errors and gives calibrated-probability metrics for free. We evaluate eight local models: a dense Qwen2.5-Instruct ladder (0.5B, 1.5B, 3B, 7B, 14B), a Qwen2.5-1.5B base model, Llama-3.2-1B-Instruct and Llama-3.1-8B-Instruct.

What we find. Every bias is far smaller than the hypotheses predict (figure 1). Frequency skew is the largest bias in small models, at +4.3 to +4.9 points for small Qwen2.5-Instruct models, +8.9 for the 1.5B base model and +16.9 for Llama-3.2-1B. It shrinks by 3.7 points per tenfold increase in parameters and is about zero at 14B (-0.6) and 8B (+0.6). Recency never produces the predicted pull. Instead, models of 3B and above are slightly *less* likely to predict the label of the final demonstrations (-1.7 to -3.6 points, Holm-corrected $p < 0.05$). Label–length correlation has no reliable effect at any size ($|\Delta_T| \leq 3.1$ points), and only 1 of 32 calibration-error differences has a confidence interval that excludes zero. Large shifts need extreme manipulations: when all ten demonstrations share one label, models of 1.5B and below shift by 16–36 points, but Qwen2.5-7B shifts by only 3.2. For every model of 3B and above, simply re-drawing a balanced demonstration set flips 4–8% of predictions. That is more than any systematic bias we measure in these models.

Our contributions are:

- We build a paired, label-rotated benchmark that measures frequency, recency, primacy and label–length biases side by side with one protocol. Every model sees the same prompts, so all comparisons are paired at the prompt level.
- We conduct a scale study over eight models and 80,000 scored prompts. It shows that frequency skew fades with scale and instruction tuning, and that recency reverses sign in models of 3B and above.
- We complement the main study with dose–response, length-interaction, calibration and resampling-noise analyses. They locate the remaining bias in small models, extreme manipulations and ambiguous tasks.
- We give practical guidance: for current instruction-tuned models of 3B and above, averaging over demonstration sets matters more than controlling label order or length.

Section 2 reviews related work. Section 3 describes the benchmark and metrics. Section 4 reports results, section 5 discusses them along with limitations, and section 6 concludes.

2 Related Work

Label biases in in-context learning. Zhao et al. [2021] identified three biases in GPT-3 few-shot classification: majority-label bias, recency bias and common-token bias. They proposed contextual calibration, which estimates the model's prior with a content-free input and divides it out. Fei et al. [2023] added domain-label bias, where the task's text domain alone favours some labels, and corrected it with domain-context calibration. Zhou et al. [2024] unified several of these methods as batch calibration. Holtzman et al. [2021] showed that the probability of a label word is also affected by competition between surface forms. These works study older base models and mostly change one factor at a time. Building on their definitions, we measure four demonstration-level statistics side by side, with one protocol, on modern instruction-tuned models. We also test whether the biases that motivated calibration are still large enough to need it.

Sensitivity to demonstration choice and order. Lu et al. [2022] showed that permuting the same demonstrations can change accuracy dramatically. Liu et al. [2022] showed that the choice of

demonstrations matters and proposed retrieving similar ones. Min et al. [2022] found that replacing gold labels with random labels hurts surprisingly little, which suggests that models use demonstrations mostly for format and label space. Unlike order-sensitivity studies, which report variance over permutations, we fix the label count and move a chosen label to the start or end. This separates directional recency and primacy effects from general instability. We also report the general instability itself, as the flip rate between two balanced demonstration sets, and use it as a noise floor.

Scale and in-context learning. Wei et al. [2023] found that larger models can override semantic priors when shown flipped labels, while smaller models cannot. This suggests that larger models pay closer attention to the demonstration labels. Our setting is different: the labels are correct, but their distribution or position is skewed. Here we find the opposite pattern for frequency skew, which larger models largely ignore. We study a dense five-point ladder within one model family, plus a base-versus-instruct pair, so that scale and instruction tuning are not confounded with model family.

Calibration of language models. Expected calibration error (ECE) [Guo et al., 2017] is the standard measure of whether a classifier's confidence matches its accuracy. Few-shot calibration work [Zhao et al., 2021, Zhou et al., 2024] usually focuses on accuracy after recalibration. We instead ask whether each bias type changes ECE and predictive entropy, and we report both for every model and condition.

3 Method

Our main question is: **when demonstrations carry a spurious statistic that favours a label T, how much more often does the model predict T?** We answer it with a paired design in which only the manipulated statistic differs between a biased prompt and its clean counterpart.

3.1 Hypotheses

We test five hypotheses, written down in our research plan before any model was run.

- **H1.** Frequency skew causes the largest bias, a 15–25 point shift toward the majority label.
- **H2.** Recency causes the second-largest bias, 8–15 points.
- **H3.** Label–length correlation causes a moderate bias (5–10 points) that does not shrink with scale.
- **H4.** Larger models (a) partly correct frequency skew but (b) amplify recency bias.
- **H5.** Expected calibration error increases under every bias type.

3.2 Tasks and Data

We use four classification tasks that differ in difficulty and label count. SST-2 [Socher et al., 2013] is binary sentiment (negative/positive). RTE and MRPC come from GLUE [Wang et al., 2018]: RTE is entailment (yes/no) and MRPC is paraphrase detection (no/yes). AG NEWS [Zhang et al., 2015] is 4-way topic classification (World/Sports/Business/Technology). For each task we draw 200 gold-balanced test items from the validation split (test split for AG NEWS). Demonstrations come from the training split and are filtered by length (SST-2: 5–60 words; RTE: $\leq$80; AG NEWS: $\leq$80; MRPC: $\leq$70). Demonstrations and test items therefore never overlap.

3.3 Demonstration Manipulations

Every prompt contains $k = 10$ demonstrations followed by the test input. Table 1 lists the conditions. The CLEAN condition has balanced label counts in random order. SKEW uses 8 demonstrations labelled T and 2 with other labels. RECENCY keeps the label counts balanced but places two T demonstrations last. PRIMACY places them first. We read "label-position correlation" at the prompt level, so PRIMACY is the mirror image of RECENCY. LENGTH keeps the counts and order balanced but draws T demonstrations from the longest third of T's pool and the others from the shortest third.

Target rotation. For every test item we build one prompt per possible target T, so each label is favoured equally often. A model with a fixed preference for, say, "positive" therefore shows no net shift toward T, and any shift we measure comes from the manipulation. With 200 items, 4 tasks, $|\mathcal{L}| \in \{2, 4\}$ targets and 5 conditions, this gives 10,000 prompts. Each (task, item, T, condition) draws its own demonstrations, seeded by an MD5 hash of the tuple, so every model sees the same

Condition	Label counts	Order	Lengths
CLEAN	balanced (5/5; 3/3/2/2)	random	random
SKEW	$8\times T$, $2\times$ other (80/20)	random	random
RECENCY	balanced	last 2 demos $= T$	random
PRIMACY	balanced	first 2 demos $= T$	random
LENGTH	balanced	random	T: longest third; others: shortest third
Supplementary dose–response set			
SKEW60 / SKEW100	$6\times T$ / $10\times T$	random	random
RECENCY4	balanced†	last 4 demos $= T$	random

Table 1: Demonstration manipulations. Every prompt has $k = 10$ demonstrations. The target label T is rotated through every label, and each (task, item, T, condition) gets its own seeded demonstration sample. In LENGTH, T demonstrations are 15–29 words longer on average than the others. †In 4-way AG NEWS, RECENCY4 raises T's share to 40%.

prompts. A supplementary set of 6,000 prompts measures dose–response: skew at 60% and 100%, and RECENCY with the last four demonstrations equal to T.

Prompt format. Prompts are raw completion text without a chat template: a one-line instruction (e.g., "Classify the sentiment of each movie review as positive or negative."), then ten `Review: ... / Sentiment: ...` pairs, then the test input followed by `Sentiment:`.

3.4 Models and Scoring

We evaluate Qwen2.5-Instruct [Qwen Team, 2024] at 0.5B, 1.5B, 3B, 7B and 14B parameters; the Qwen2.5-1.5B base model, to isolate instruction tuning; and Llama-3.2-1B-Instruct and Llama-3.1-8B-Instruct [Grattafiori et al., 2024]. All models run locally with vLLM [Kwon et al., 2023] in bf16 with greedy decoding. The GPUs were shared with other jobs, so the 7B and 8B models use 4-bit AWQ checkpoints [Lin et al., 2024].

We score each prompt with the next-token distribution restricted to the first token of each label word (with a leading space), renormalised over the label set. The prediction is the argmax. This removes format and parsing failures, and it gives a probability vector $\boldsymbol{p} \in \Delta^{|\mathcal{L}|}$ for the calibration metrics.

3.5 Metrics

Target shift. Our primary metric is the shift toward the target label,

$$\Delta_T(c) = \Pr[\hat{y} = T \mid c] - \Pr[\hat{y} = T \mid \text{CLEAN}], \tag{1}$$

paired on (task, item, T) and reported in percentage points. We average over targets and items within a task, then weight the four tasks equally. Positive values mean the model follows the spurious statistic.

Secondary metrics. We report accuracy, overall and split by whether the gold label equals T. We report ECE [Guo et al., 2017] with 10 equal-width bins on max-probability confidence, computed per task and averaged. We also report predictive entropy normalised by $\log|\mathcal{L}|$, and the mean probability mass on T.

Noise floor. To judge whether a bias matters in practice, we compare it to the flip rate between predictions under two independently drawn CLEAN demonstration sets for the same item.

Statistics. We compute 95% confidence intervals (CIs) with a cluster bootstrap over test items, resampled within each task (2,000 resamples; 1,000 paired resamples for ΔECE). Significance comes from sign-flip permutation tests on per-item Δ_T, with Holm correction [Holm, 1979] over all 32 (model $\times$ condition) tests. For scale trends we fit an OLS slope of Δ_T on $\log_{10}$(parameters) along the Qwen2.5-Instruct ladder, with a bootstrap CI, and report Spearman's ρ across the five sizes.

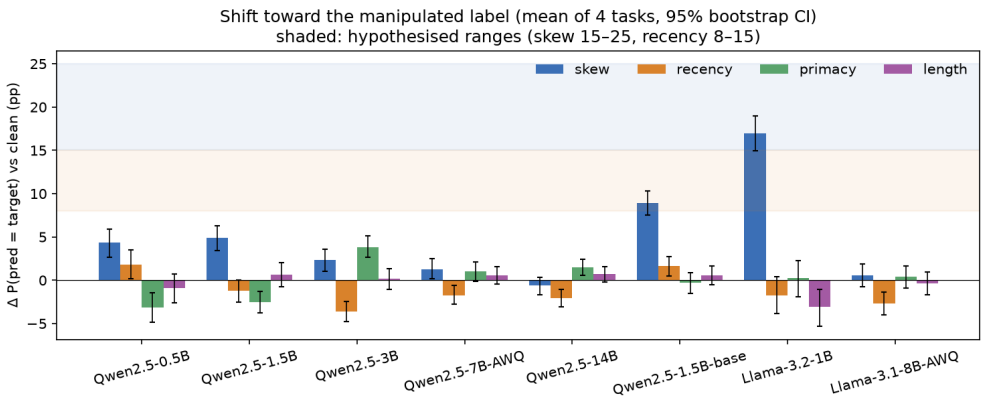


Figure 1: Shift toward the manipulated label T (4-task mean, 95% bootstrap CIs). The shaded bands mark the hypothesised ranges for skew (15–25 points, blue) and recency (8–15 points, orange). Only Llama-3.2-1B under frequency skew reaches its band. Recency is negative for every model of 3B and above, and length effects stay within ± 3 points.

Model	SKEW (80/20)	RECENCY	PRIMACY	LENGTH
Qwen2.5-Instruct ladder				
Qwen2.5-0.5B	**+4.3*** [+2.7, +5.9]	+1.8 [+0.2, +3.5]	−3.1* [−4.8, −1.5]	−0.9 [−2.6, +0.7]
Qwen2.5-1.5B	**+4.9*** [+3.4, +6.3]	−1.2 [−2.5, −0.0]	−2.5* [−3.8, −1.3]	+0.7 [−0.7, +2.0]
Qwen2.5-3B	+2.3* [+1.1, +3.6]	−3.6* [−4.8, −2.4]	**+3.8*** [+2.7, +5.1]	+0.2 [−1.0, +1.3]
Qwen2.5-7B-AWQ	**+1.3** [+0.2, +2.5]	−1.7* [−2.8, −0.6]	+1.0 [−0.1, +2.1]	+0.6 [−0.4, +1.6]
Qwen2.5-14B	−0.6 [−1.7, +0.3]	−2.1* [−3.1, −1.1]	**+1.5*** [+0.6, +2.4]	+0.7 [−0.2, +1.6]
Base model and second family				
Qwen2.5-1.5B-base	**+8.9*** [+7.6, +10.3]	+1.6 [+0.5, +2.7]	−0.3 [−1.5, +0.8]	+0.5 [−0.5, +1.6]
Llama-3.2-1B	**+16.9*** [+14.9, +19.0]	−1.8 [−3.8, +0.4]	+0.2 [−1.9, +2.3]	−3.1 [−5.3, −1.1]
Llama-3.1-8B-AWQ	**+0.6** [−0.8, +1.8]	−2.7* [−4.0, −1.3]	+0.4 [−0.9, +1.7]	−0.3 [−1.7, +1.0]
Hypothesised	+15 to +25	+8 to +15	—	+5 to +10

Table 2: Target shift Δ_T (percentage points) toward the manipulated label, as an equal-weight mean over four tasks, with 95% cluster-bootstrap CIs. *: Holm-adjusted $p < 0.05$ over 32 tests. **Bold**: largest positive shift per model. Only Llama-3.2-1B under SKEW reaches the hypothesised range. Recency is negative, not positive, for every model of 3B and above.

4 Results

We first report the size of each bias (section 4.1), then how it changes with scale (section 4.2) and with the strength of the manipulation (section 4.3). We close with calibration, the length interaction and the noise floor (section 4.4).

4.1 Every Bias Is Much Smaller Than Hypothesised

Figure 1 and table 2 show the main result. **Frequency skew** is the largest bias in every model of 1.5B or smaller: +4.3 and +4.9 points for the small Qwen2.5-Instruct models, +8.9 for the 1.5B base model and +16.9 for Llama-3.2-1B. Only Llama-3.2-1B falls inside the hypothesised 15–25 point range. At 14B and 8B, skew has no detectable effect (-0.6 and +0.6).

Recency never reaches the predicted 8–15 points. Its largest positive value is +1.8 (Qwen2.5-0.5B), which does not survive Holm correction. From 3B upward the sign flips: two final T demonstrations make the model *less* likely to predict T, by -1.7 to -3.6 points. This holds for all four models of 3B and above, and it is significant after correction in each. **Primacy** roughly mirrors this pattern: it is negative for 0.5B and 1.5B (-3.1, -2.5) and positive at 3B (+3.8) and 14B (+1.5). **Label–length correlation** has no reliable effect. The largest magnitude is -3.1 points (Llama-3.2-1B), in the direction opposite to the hypothesis, and no length effect survives correction.

Model	Δ_T under SKEW (pp)				Flip rate (%) MRPC/mean
	SST-2	RTE	AG NEWS	MRPC	
Qwen2.5-0.5B	−1.2	−3.2	−0.1	**+22.0**	23.0 / 13.1
Qwen2.5-1.5B	−1.5	0.0	+3.8	**+17.2**	23.5 / 9.4
Qwen2.5-3B	+3.0	−0.5	+1.8	**+5.0**	11.5 / 6.1
Qwen2.5-7B-AWQ	+1.0	−1.8	+0.9	**+5.0**	8.5 / 4.9
Qwen2.5-14B	0.0	−0.5	**+0.8**	−2.8	9.0 / 4.2
Qwen2.5-1.5B-base	−1.5	+7.8	+9.8	**+19.8**	6.0 / 6.2
Llama-3.2-1B	+3.8	+22.2	0.0	**+41.8**	41.5 / 22.8
Llama-3.1-8B-AWQ	−0.8	+2.5	−5.1	**+5.8**	14.5 / 7.8
Clean acc. range (%)	89.8–95.5	64.2–92.5	69.6–85.5	54.5–77.0	

Table 3: Skew bias by task, and the noise floor. **Bold**: task with the largest skew shift per model. SST-2 is essentially immune, while MRPC, the task with the lowest clean accuracy, carries most of the bias. The last column is the share of predictions that flip when one balanced CLEAN demonstration set is replaced by another, on MRPC and averaged over tasks.

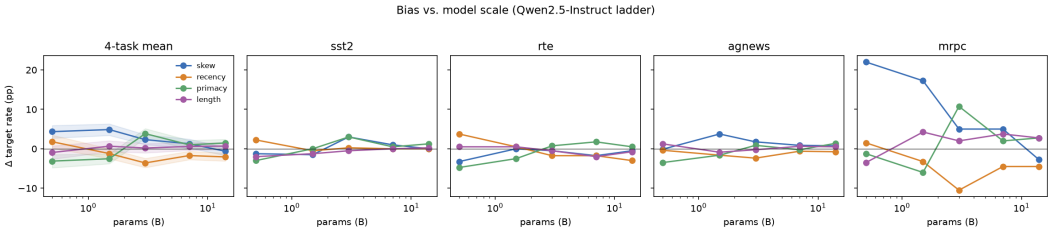


Figure 2: Target shift against parameter count along the Qwen2.5-Instruct ladder. Left: 4-task mean with a 95% CI band. Right: per task. Skew (blue) falls with scale, driven almost entirely by MRPC. Order effects (recency, orange; primacy, green) are non-monotone and largest at 3B, where MRPC recency reaches -10.5 points.

Bias lives in the ambiguous task. Table 3 splits the skew effect by task. SST-2, where clean accuracy is 90–96% for every model, is essentially immune ($|\Delta_T| \leq 3.8$). MRPC has the lowest clean accuracy and the highest resampling flip rate, and it carries most of the bias: +17 to +22 points for the small Qwen2.5-Instruct models, +19.8 for the base model and +41.8 for Llama-3.2-1B. The 4-task averages in table 2 therefore understate the risk on hard tasks and overstate it on easy ones.

Where skew hurts. Because T is rotated, the accuracy that skew adds on items whose gold label is T cancels the accuracy it removes on the others. Overall accuracy therefore changes by at most 2.8 points under any condition (figure 4, right). The split by gold label shows the real cost. For Llama-3.2-1B, skew lowers accuracy on minority-class items (gold $\neq T$) from 73.3% to 58.8% ↓and raises it on majority-class items from 72.0% to 86.2% ↑. The 1.5B base model shows the same pattern (79.8% → 70.1% and 79.2% → 88.4%). Qwen2.5-14B is unaffected (87.2% → 87.7% and 87.1% → 86.9%).

4.2 Scale Removes Skew and Turns Recency Negative

Figure 2 and table 4 show how each bias changes along the Qwen2.5-Instruct ladder. Skew falls by 3.7 points per tenfold increase in parameters ($\rho = -0.9$). The Llama pair agrees: skew drops from +16.9 at 1B to +0.6 at 8B. Recency also falls with scale (-2.4 points per decade), but it falls *below zero* rather than rising as H4 predicted. The Llama pair again agrees (-1.8 at 1B, n.s.; -2.7 at 8B, significant). Primacy rises with scale (+3.6 points per decade). Order effects are not monotone, however: the 3B model shows the largest recency (-3.6) and primacy (+3.8) effects of the ladder. The length slope is not distinguishable from zero.

Instruction tuning. At a fixed size of 1.5B, the base model is about twice as sensitive to skew as its instruction-tuned sibling (+8.9 vs. +4.9 points). It is also the only model besides 0.5B with a positive recency point estimate (+1.6).

Bias	Slope (pp / decade)	95% CI	Spearman ρ
SKEW	-3.7	$[-4.8, -2.4]$	-0.9
RECENCY	-2.4	$[-3.6, -1.2]$	-0.7
PRIMACY	$+3.6$	$[+2.4, +4.7]$	$+0.7$
LENGTH	$+1.0$	$[-0.2, +2.2]$	$+0.7$

Table 4: OLS slope of Δ_T on $\log_{10}$(parameters) along the Qwen2.5-Instruct ladder (five sizes, 0.5B–14B). The CIs reflect item-sampling noise only, not model-to-model variation.

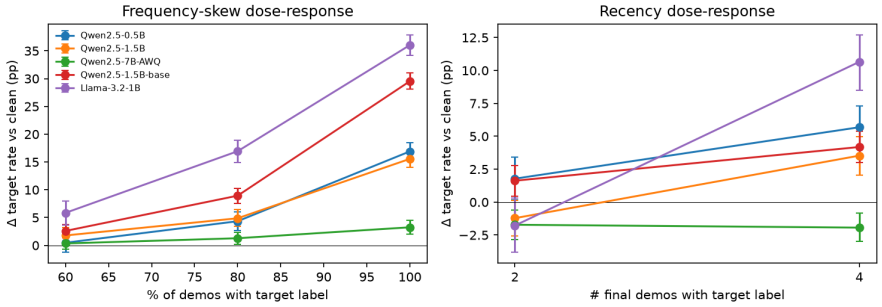


Figure 3: Dose–response on five models. Left: frequency skew at 60%, 80% and 100% target share. Right: recency with the last two or four demonstrations equal to T. Skew bias grows sharply once the minority label disappears from the context. Qwen2.5-7B (green) stays nearly flat under both manipulations.

4.3 Large Shifts Require Extreme Manipulations

Figure 3 and table 5 report the supplementary dose–response set. Skew bias is strongly non-linear. At 60% target share it is at most +5.8 points; at 80% it reaches +4 to +17; when all ten demonstrations share one label it jumps to +15.6 to +36.0 points in the small models. The base model again roughly doubles its instruction-tuned sibling (+29.5 vs. +15.6). Most of the skew effect thus appears only when the minority label is (nearly) absent from the context.

Recency behaves differently with longer runs. A run of four final T demonstrations gives a positive recency effect in all small models (+3.5 to +10.7), including 1.5B and Llama-3.2-1B, where a run of two had a negative point estimate. **Qwen2.5-7B is almost immune to both extreme doses.** All-T demonstrations move it by only +3.2 points, and four final T demonstrations still produce anti-recency (-1.9). Scale therefore removes not only the moderate biases but most of the dose–response.

4.4 Calibration, Length Interaction and Noise Floor

Calibration does not degrade. Table 6 and figure 4 show that ΔECE ranges from -1.3 to +2.2 points. Nine of the 32 point estimates are negative. Only one CI excludes zero (Qwen2.5-7B, PRIMACY: +1.2 [+0.02, +2.3]), and only before correcting for multiple comparisons. The largest point estimate, Qwen2.5-1.5B under skew (+2.2 [-0.05, +3.3]), does not. Normalised entropy barely moves (e.g., 0.099 vs. 0.102 for 14B under skew). The larger difference is between models: Qwen2.5-Instruct models of 3B and above are overconfident even on CLEAN prompts (ECE 0.10–0.12, against 0.05–0.07 for 0.5B and 1.5B). This is a property of the models, not of the manipulations.

No length rule. If the LENGTH condition taught a "long input $\rightarrow T$" rule, the shift would be larger for long test inputs (above the task median) than for short ones. The long-minus-short difference ranges from -0.6 to +1.9 points, and every 95% CI contains zero. The models did not learn a length-to-label association, even though T demonstrations were 15–29 words longer on average.

The noise floor dominates in larger models. Re-drawing a balanced CLEAN demonstration set flips 4.2% (Qwen2.5-14B) to 22.8% (Llama-3.2-1B) of predictions, and up to 41.5% on MRPC

	Frequency skew (share of T)			Recency (final demos $= T$)	
Model	60%	80%	100%	last 2	last 4
Qwen2.5-0.5B	+0.5 [−1.2, +2.0]	+4.3	+16.9 [+15.4, +18.5]	+1.8	+5.7 [+4.2, +7.3]
Qwen2.5-1.5B	+1.8 [+0.5, +3.2]	+4.9	+15.6 [+14.0, +17.2]	−1.2	+3.5 [+2.0, +5.0]
Qwen2.5-1.5B-base	+2.6 [+1.5, +3.8]	+8.9	+29.5 [+28.0, +31.0]	+1.6	+4.2 [+3.0, +5.4]
Llama-3.2-1B	+5.8 [+3.7, +7.9]	+16.9	**+36.0** [+34.1, +38.0]	−1.8	**+10.7** [+8.5, +12.7]
Qwen2.5-7B-AWQ	+0.3 [−0.7, +1.4]	+1.3	+3.2 [+2.0, +4.5]	−1.7	−1.9 [−3.0, −0.8]

Table 5: Dose–response of target shift Δ_T (pp, 4-task mean, 95% CI). The 80% and last-2 columns repeat table 2. Skew bias grows sharply once the minority label disappears from the context. A run of four final T demonstrations produces positive recency in small models but not in Qwen2.5-7B. **Bold**: largest shift per column group.

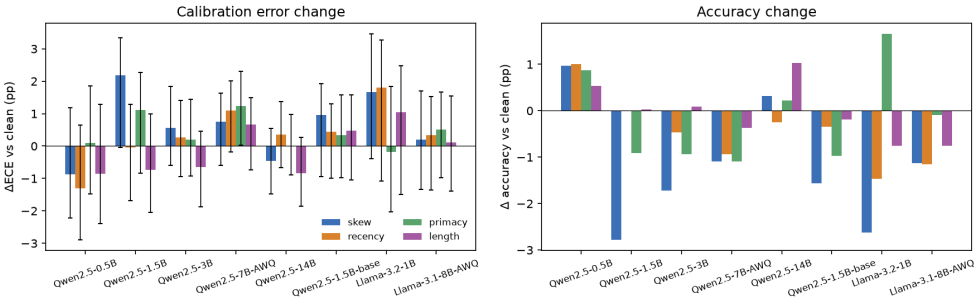


Figure 4: Left: change in ECE against CLEAN, with paired bootstrap 95% CIs. Right: change in overall accuracy. Only one of 32 ΔECE intervals excludes zero, and overall accuracy moves by less than 3 points, because target rotation makes the gains and losses cancel.

for Llama-3.2-1B (table 3). For every model of 3B and above, this ordinary resampling instability (4.2–7.8%) is larger than any systematic bias in table 2 (at most 3.8 points).

Summary. Table 7 collects the verdicts. Only the claim that scale corrects frequency skew is fully supported. The claim that skew is the largest bias holds for small models, but its magnitude is far below the prediction for instruction-tuned models.

5 Discussion

Why are the biases so small? We see three contributing factors. First, **instruction tuning and scale weaken the demonstration prior.** At 1.5B, the base model is twice as sensitive to skew as its instruction-tuned sibling, both at 80% (+8.9 vs. +4.9) and at 100% (+29.5 vs. +15.6). The large effects reported for GPT-3 [Zhao et al., 2021] come from base models, which is consistent with this gap. Our design cannot separate model era from training recipe, however. Second, **bias appears where the model is uncertain.** On SST-2, where most models are above 90% accurate, no manipulation moves predictions. On MRPC, where accuracy is lowest and predictions are least stable, skew moves small models by 17–42 points. A simple account fits both observations: the demonstration label distribution acts as a prior, and this prior only wins when the evidence from the input is weak. As models become more competent and more confident (normalised entropy falls from 0.43–0.65 at $\leq$1.5B to about 0.10 for Qwen at 7B and 14B), the prior has less room to act. Third, the four tasks are public and probably seen in pretraining, which may make models rely less on demonstrations than they would on a new task.

Why anti-recency? For models of 3B and above, two final T demonstrations make T *less* likely, and primacy shows the mirror effect. We see two possible explanations. The model may expect labels to alternate, a gambler's-fallacy-like prior learned from roughly balanced data. Alternatively, this is a side effect of our design: with balanced counts, placing two T demonstrations at the end leaves the middle positions slightly short of T (only 38% of the first two slots are T). With four final T demonstrations the effect turns positive in small models but stays negative in Qwen2.5-7B.

	ECE					Norm. entropy	
Model	CLEAN	SKEW	RECENCY	PRIMACY	LENGTH	CLEAN	SKEW
Qwen2.5-0.5B	.071	.062	.058	.072	.062	.575	.569
Qwen2.5-1.5B	.049	.070	.048	.060	.041	.427	.433
Qwen2.5-3B	.121	.127	.124	.123	.114	.170	.185
Qwen2.5-7B-AWQ	.111	.118	.121	.123†	.117	.105	.118
Qwen2.5-14B	.097	.092	.101	.097	.089	.099	.102
Qwen2.5-1.5B-base	.058	.067	.062	.061	.063	.540	.537
Llama-3.2-1B	.035	.051	.053	.033	.045	.651	.645
Llama-3.1-8B-AWQ	.053	.055	.056	.058	.055	.337	.357

Table 6: Expected calibration error (equal-task mean) and normalised predictive entropy. ΔECE against CLEAN lies between -1.3 and $+2.2$ points. 9 of 32 differences are negative, and only one paired bootstrap CI excludes zero (†: +1.2 [+0.02, +2.3], before correcting for 32 comparisons). Differences between models (larger Qwen models are overconfident even on CLEAN prompts) are much larger than differences between conditions.

	Prediction	**Result**	**Verdict**
H1	skew largest, 15–25 pp	largest in all models $\leq$1.5B (4.3–16.9 pp); only Llama-3.2-1B in range; $\approx$0 at 8–14B	partly supported
H2	recency second, 8–15 pp	$\leq$+1.8 pp; significantly negative (-1.7 to -3.6) at $\geq$3B	refuted
H3	length 5–10 pp, scale-resistant	$\lvert\Delta_T\rvert \leq 3.1$, none significant; no length rule	refuted
H4a	scale corrects skew	-3.7 pp/decade; Llama 16.9 $\rightarrow$ 0.6	supported
H4b	scale amplifies recency	recency turns negative with scale	refuted
H5	ECE rises under every bias	1/32 CIs above 0 (uncorrected); 9/32 negative	refuted

Table 7: Verdicts on the five hypotheses.

The recency bias described in the earlier literature therefore appears only when the run is long or the model is small. Separating the two explanations requires varying run length and alternation structure directly, which we leave to future work.

Length is not a label cue. Even with a 15–29 word gap between T and non-T demonstrations, no model learned to map length to label, either on average or on long test inputs. This fits the view of Min et al. [2022] that models use demonstrations mostly for label space and format, rather than for fine-grained input–label statistics.

Practical implications. For instruction-tuned models of 3B and above on standard tasks, the specific labels of a balanced-looking demonstration set matter less than ordinary demonstration-to-demonstration variance. Practitioners will gain more from averaging over several demonstration sets, or from choosing good demonstrations, than from tuning label order or matching input lengths. Frequency skew still matters in three cases: small models, base models and ambiguous tasks. In these cases, balancing labels or applying contextual calibration [Zhao et al., 2021, Fei et al., 2023] remains worthwhile, and demonstration sets with a single label should be avoided.

5.1 Limitations

Model coverage. We study one dense ladder (Qwen2.5-Instruct) plus two Llama models. The 7B and 8B models are 4-bit AWQ checkpoints, so quantisation is confounded with size at those points. Our largest model has 14B parameters; we do not know whether the trends continue to 70B and beyond.

Scale statistics. Slope CIs reflect item-sampling noise only. With five sizes, the evidence about variation across models is weak, and the Spearman correlations rest on $n = 5$.

Design choices. We implemented label-position correlation as prompt-level primacy. RECENCY and PRIMACY slightly unbalance the labels in the middle positions, and in AG NEWS, RECENCY4

raises T's share to 40%. We used a single raw-completion template with $k = 10$ and fixed label words; chat-formatted prompts or other values of k may behave differently.

Sample size and data. With 200 test items per task, the CIs on the 4-task mean are about $\pm$1–2 points, so effects below about 2 points are hard to resolve. All four tasks are public benchmarks that are probably in pretraining data, which may deflate bias estimates relative to new tasks. The dose–response set covers only five models (not 3B, 14B or Llama-3.1-8B) because of limited GPU time.

Broader impact. This work measures robustness and does not introduce new capabilities. Its main risk is over-generalisation: our null results for modern models do not show that few-shot classifiers are safe from demonstration bias on new, ambiguous or high-stakes tasks, where we find the largest effects.

6 Conclusion

We measured how four spurious statistics in few-shot demonstrations (frequency skew, recency, primacy and label–length correlation) shift the predictions of eight local LLMs, using 80,000 paired, label-rotated prompts. All four biases are far smaller than hypothesised. Frequency skew is the only consistently harmful one. It matters mainly for small models, base models and ambiguous tasks, and it fades with scale at 3.7 points per tenfold increase in parameters. Recency, primacy and length produce effects of at most 4 points, including an unexpected anti-recency pattern in models of 3B and above. Calibration is essentially unaffected. The key takeaway is that, for current instruction-tuned models of 3B and above, ordinary variation between balanced demonstration sets (4–8% of predictions flip) outweighs any systematic label bias we found.

Future work should extend the study to 32B–70B models and chat-formatted prompts. It should also use new or synthetic tasks with randomised label words, to remove prior knowledge and contamination. Varying run length and label alternation would test the alternation-expectation account of anti-recency. Finally, it remains open whether contextual [Zhao et al., 2021] or domain-context [Fei et al., 2023] calibration still helps when the biases are this small.

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A Experimental Details

Data sources. SST-2 and AG NEWS were loaded from the local Hugging Face cache. RTE and MRPC were loaded from the `nyu-mll/glue` parquet files. Test sets are gold-balanced: 100/100 for binary tasks and 50 per class for AG NEWS.

Example prompt (SST-2).

```
Classify the sentiment of each movie review as positive or negative.

Review: <demo text>
Sentiment: positive
```

```
... (10 demonstrations) ...
Review: <test text>
Sentiment:
```

Scoring. We use vLLM 0.32 with `allowed_token_ids` set to the first token of each label word (with a leading space) and `logprobs_mode="processed_logprobs"`, so the returned log-probabilities are renormalised over the label set. Decoding is greedy with seed 0.

Seeds. Demonstration sampling is seeded by an MD5 hash of (task, item, T, condition). The analysis RNG seed is 0.

Compute. Each run used one slice of a shared NVIDIA RTX A6000 (48 GB), with 5–44 GB available depending on other users' jobs. The 14B run took 74 minutes; the 0.5B–1.5B runs took 2–4 minutes each. The total was about 6 GPU-hours, with no API cost.

Software. vLLM 0.32.0, PyTorch 2.13, Transformers 5.19, pandas 3.0, matplotlib, Python 3.12.

Length-interaction results. Table 8 lists the difference in LENGTH shift between long and short test inputs.

Model	Short inputs	Long inputs	Long − short [95% CI]
Qwen2.5-0.5B	−1.1	+0.2	+1.4 [−2.2, +4.2]
Qwen2.5-1.5B	0.0	+0.7	+0.7 [−2.0, +3.5]
Qwen2.5-3B	+0.4	−0.2	−0.6 [−2.8, +2.3]
Qwen2.5-7B-AWQ	+0.1	+1.2	+1.1 [−1.2, +2.9]
Qwen2.5-14B	+0.6	+0.7	+0.2 [−1.8, +1.9]
Qwen2.5-1.5B-base	−0.1	+1.6	+1.7 [−0.5, +3.8]
Llama-3.2-1B	−3.4	−1.6	+1.8 [−2.7, +5.6]
Llama-3.1-8B-AWQ	−1.1	+0.7	+1.9 [−0.3, +4.7]

Table 8: LENGTH target shift (pp) for test inputs below and above the task median length. Every CI contains zero, so we find no evidence of a learned length-to-label rule.